AI-SANA NATIONAL AI TRAINING INITIATIVE

PROGRAM 2

SYLLABUS

Digital Solutions to Industry Challenges through AI and Team Leadership

Format:	Online (via LMS Platonus + SMILE)
Duration:	16 weeks (3.5 months)
Total hours (compulsory):	235,5 hours
Total hours (with optional):	279,86 hours
Video content:	74,86 hrs (30,5 required + 60,84 optional)
Weekly commitment:	Approximately 15–17.5 hours/week
Target enrollment:	Up to 60,000 students
Assessment:	SMILE-based + Capstone Project
Grading:	Pass / No Pass (70% threshold)

Hours Breakdown:

Video lectures (required):	30,5 hours
Video lectures (optional):	60,84 hours
Reading and research:	60 hours
Practical experiments:	60 hours
Team interaction:	15 hours
Exam assignments:	60 hours
Capstone Project – The Final Stage:	10.5 hours. Video instructions: 33,08 min + presentation format + evaluation criteria. Project preparation: 10 hrs
TOTAL (compulsory):	**235,5 hours**
Total compulsory + optional:	**279,86 hours**

1. General information

Program name	AI-SANA Program 2: Digital Solutions to Industry Challenges through AI and Team Leadership
Program level	Professional development / National training initiative
Duration	16 weeks (3.5 months), 253,92 compulsory hours + 38,02 optional hours
Weekly commitment	Approximately 15–17.5 hours per week
Delivery format	Fully online via LMS (Platonus) with SMILE integration
Target enrollment	Up to 60,000 students
Number of modules	6 modules (2.1–2.5 plus Capstone The final stage Module 2.6)
Assessment	Pass / No Pass - 70% threshold on all SMILE activities

Prerequisites	Successful completion of AI-SANA Program 1 Intellectual Technologies and Entrepreneurial Thinking program (or equivalent). Basic Python programming experience and familiarity with machine learning fundamentals. Comfort with Google Colab and GenAI tools.
Postrequisites	None. Program 2 is the advanced capstone program of the AI-SANA National Training Initiative.
International partners: 34	**Core faculty and expert practitioners:** • **Paul Kim**, PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank • **Nhi Le** - Principal at Alpha Intelligence Capital (AIC) - a $500M global venture capital fund dedicated exclusively to deep-tech AI and machine-learning companies, investing from Seed to Series B. Nhi leads sourcing and evaluation of new deals across the AI sector and serves as Board Observer at Exostellar (the AI infrastructure orchestration platform she backed at Series A in 2023, acquired by Qualcomm) and Board Member at Marro Technologies. In collaboration with Carnegie Mellon University's Swartz Center for Entrepreneurship, Nhi co-developed the newly launched Deep Tech Venture-Ready Program - an 18-month initiative backed by $240M in soft-circled capital from 30 partner VC firms that prepares CMU faculty, PhDs, and alumni to commercialize breakthrough science across AI, robotics, advanced materials, life sciences, and energy. She also co-hosts AIC's immersive Deep Tech VC 101 Bootcamps at CMU and in Europe, and serves on the Project Olympus advisory board. • **Nick Kelley** - Venture Advisor / Partner at Alpha Intelligence Capital (AIC) - a $500M global venture capital fund dedicated exclusively to deep-tech AI and machine-learning companies, investing from Seed to Series B. Based in Basel, Switzerland. Previously Director of Data Science & A.I. at Novartis, where he led enterprise-scale AI initiatives inside one of the world's largest pharmaceutical companies - giving him a rare hands-on view of how AI actually gets deployed, validated, and scaled inside highly regulated life-science organizations. A.I. Strategy and Deep Tech Advisor at Blue Zebra. Collaborating with MIT and Microsoft Research • **Alex Aguilar** - Associate and Team Lead for Mining at Mind the Bridge a global open innovation platform headquartered in Silicon Valley that connects Fortune 500 corporates with startups. Leads the mining innovation vertical, helping companies in mining, defense, shipbuilding, and energy identify and integrate emerging technologies. Over 8 years of experience in business development and innovation across LATAM, Asia, and Europe. Previously led B2B business development at Huawei in Latin America and held senior roles at Linio, Sodimac, and Grupo Pacific Traders. At Mind the Bridge has supported Fortune 500 energy and cleantech corporates - including Enel, ENI, and Terna - with startup scouting, M&A sourcing, benchmarking, and market landscaping. Holds an executive education at MIT. • **Michael Moe** - Founder and CEO at GSV Holdings. Co-founder of the ASU GSV Summit. The world's largest event at the intersection of

education, edtech, workforce skills, and venture capital. Forbes calls it the "Davos of Education," and The New York Times describes it as the "must-attend event for education technology." Founded in 2010 in partnership with Arizona State University and GSV Ventures (Global Silicon Valley). Founder of GSV University and GSV MBA. Author of books: “Finding the Next Starbucks”, “The Global Silicon Valley Handbook,” “The Mission Corporation...Contemporary Capitalism”.

- **Guilianna Crivello** - Founder of Stealth Startup, a network for entrepreneurs building in stealth and a venture partner at OFF PISTE, a venture studio operating at the intersection of thematic investing, control/ownership models, and vertical AI implementation across private equity and startups. Previously Founding General Partner and Head of Ventures at Draper Startup House Ventures (part of the Draper Venture Network, - launched the firm's first global micro-fund in 2021 together with Vikram Bharati, investing in early-stage founders across 60+ countries and 30+ industries, with strong exposure to fintech deal flow in emerging markets across Southeast Asia, India, Latin America, and Eastern Europe. Made 200+ investor introductions to global founders and led the firm's transition from a hospitality-first startup platform into a true cross-border venture fund during the pandemic.
- **Tahira Dosani** - General Partner and Co-Founder of ResilienceVC. Adjunct Professor at The Johns Hopkins University - Paul H. Nitze School of Advanced International Studies (SAIS). Adjunct Professor at Georgetown University
- **Axel Badalian** - Principal at Alpha Intelligence Capital (AIC) - a $500M global venture capital fund dedicated exclusively to deep-tech AI and machine-learning companies, investing from Seed to Series B with a globally distributed team of deep-tech entrepreneurs, scientists, and investors. Based in Paris, Axel leads sourcing and investment execution across the AI/ML enabling stack - from compute, hardware, and infrastructure to foundation models and verticalized applications. He is also listed as an AI Expert at OECD.AI, contributing to policy discussions on AI computing capacity, infrastructure scale-up, and its environmental impact.
- **Arun Vijay -** Investment Analyst at Alpha Intelligence Capital (AIC) - a $500M global venture capital fund dedicated exclusively to deep-tech AI and machine-learning companies, investing from Seed to Series B out of offices in San Francisco and Singapore. At AIC, Arun supports sourcing and due diligence across the AI/ML enabling stack, with direct exposure to the cybersecurity portfolio - a core thesis area for the fund, which includes investments in ZeroEyes (AI-powered gun detection for public and private spaces), X-37 (AI-driven cyber defense), and infrastructure-security plays like Exostellar (secure AI workload orchestration across heterogeneous hardware - acquired by Qualcomm). These investments sit at the center of the defining cybersecurity questions of the AI era: how to defend AI systems themselves, how to use AI to detect and respond to threats at machine speed, and how to secure the compute and data infrastructure on which modern enterprises now depend..In parallel, he is building OpenMentor.AI, a community-driven mentorship platform for students.
- **David Forsberg -** Managing Partner and Founder of Ascent Energy Ventures - a Denver-based specialist venture capital firm focused on the digital transformation of the energy industry. Ascent invests in startups building the software, automation, AI, and data infrastructure layer of the global energy stack - from modernizing legacy SCADA systems in oil & gas, to regulatory intelligence tooling, grid-planning software, and digital

platforms across power, utilities, and hydrocarbons. Ascent's thesis is explicitly pragmatic: rather than picking a side between hydrocarbons and decarbonization, the firm backs the founders building the automation and digital tools that make all forms of energy production more efficient, transparent, and measurable. Also serves as a strategic advisor to energy founders and private investors

- **Charlie Kleinsmith** - Principal at SineWave Ventures - a $160M early-stage venture capital firm with offices in New York and San Francisco, dedicated to backing technologies that bridge commercial enterprise and public-sector adoption. Charlie leads investments at the intersection of data, AI, cybersecurity, networking, and compute infrastructure - precisely the stack underpinning modern operational automation. Publishes thought leadership on data provenance, zero-trust security architectures, and data control frameworks, which are central to deploying automation inside regulated enterprises and government agencies. SineWave's partnership bench itself - former leaders from NIST (cybersecurity program lead), CIA, NSA, Department of Commerce, GSA, and DoD.
- **Meïssa Gueye** - Head of Venture Capital Africa at IFC (International Finance Corporation) a member of the World Bank Group and the largest global development institution focused exclusively on the private sector in emerging markets, with a record $71.7 billion committed in fiscal year 2025. Meïssa leads IFC's equity and fund investments into African tech and growth-stage companies, deploying capital directly and via VC funds and accelerators across Africa. IFC is the largest emerging-market fintech investor globally, with an early-stage fintech portfolio exceeding $600M and over $3B committed across 175+ tech companies and 120+ VC funds. Based in Dakar, Meïssa sits at the center of the African venture ecosystem - frequently cited as one of the investors every African founder should know.
- **Kieran Gartlan** - Managing Partner at the Yield Lab LATAM, one of Latin America's leading AgriFood and Climate Tech venture capital firms, with offices in Argentina, Brazil, and Chile, and part of The Yield Lab global network with regional funds across North America, Europe, Latin America, and Asia Pacific. Kieran leads a portfolio spanning the full farm-to-fork spectrum - with access to 1,100+ startups across Latin America. Board Member at Produzindo Certo, BemAgro S.A, Grão Direto, SEEDZ. Winner of Jesse H Neal Award 2010 for best recurring Webcast covering South American Grains Market. Winner of National Excellence Award 2007 from the NAAJ (North American Agricultural Journalists Association) for analysis of Brazil’s grain sector. Author of "Cereal Killers – Grain Markets 101” commissioned by U.K publisher Harriman House
- **Kevin Dowling** - Managing Director of the Robotics Factory and Hardware at Innovation Works - Pittsburgh's flagship startup program, part of a $63M U.S. Economic Development Administration grant to grow the robotics and AI startup ecosystem across the region that is widely recognized as the "Robotics Capital of the World," with 100+ robotics companies clustered there. Kevin leads the Factory's three interconnected programs - Create, Accelerate, and Scale - which work specifically with sectors including construction, aerospace, mining, and defense to identify industry problems and turn them into venture-backed robotics startups. Also a founding member and Board Chair of the Pittsburgh Robotics Network, the regional industry body coordinating talent, capital, and policy for the sector.Holds a PhD and spent 17 years as Scientist and Senior Engineer at Carnegie Mellon University's Robotics Institute, one of

	the most influential robotics research centers in the world. Former VP of Color Kinetics (acquired by Philips), where his work on intelligent LED lighting systems generated one of the most valuable patent portfolios in the industry. • **Bola Adegbulu** - AI Consultant at Infinite Ascent Ventures.Ex-Principal in AI Fund and acceleration program, led by Prof. Andrew Ng's. Board Member at Africa Deep Tech Community. Member of PEF Community. Europe 30 under 30: Manufacturing & Industry. • **Peter Turner -** Partner at Breakthrough Energy Ventures. BEV invests in companies that leverage innovative technologies to help address climate change. Backed by many of the world's top business leaders, BEV has more than $2 billion in committed capital to support bold entrepreneurs building companies that can significantly reduce emissions from agriculture, buildings, electricity, manufacturing, and transportation. EVP Selection Committee in Stanford Woods Institute for the Environment • **Seamus McAteer** - Chief Executive Officer at Speechlab. Non Executive Director at On Device Research. Founder in residence in AI fund (Managed by Andrew Ng). Investor & Advisor. 3 exits. • **Alberto Pellicione** - Partner at Alpha Intelligence Capital. AI Investor. Ex Global XDR Strategy Leader at IBM, participated in developing QRadar+ XDR strategy on a global scale. Serial Entrepreneur in Cybersecurity field. • **Hyu Dong** - NUS Senior Sales Specialist at Nutanix, a global leader in cloud software, offering organizations a single platform for running apps and data across clouds, with capitalisation more than 10 bln dollars. 3 IPO/Acquisitions • **Tom Pereira** - Chief Financial Officer at at Advanced Navigation, an Australian unicorn, a global leader in navigation and autonomous systems which serves over 2800 clients across 70+ countries (among clients Boeing, Airbus, NASA and others) • **Scott Nuzum** - Special Counsel at Wiley Rein Attorney at Wiley Rein LLP (Washington, D.C.) - advising dual-use and deep tech companies on corporate law, national security, and emerging technology regulation: M&A, cybersecurity compliance, and technology licensing. Works with founders and venture capital funds across the defense, energy, space, and AI sectors on cross-border deals. Over 15 years of experience at the intersection of law, business, and public policy. Previously held legal and policy positions in the US federal government, served as General Manager at the venture studio Innovyz USA, co-founded a data marketplace for agtech applications. • **Scott Felder -** Partner at Wiley Rein LLP (Washington, D.C.) and head of the firm's Patent Prosecution practice. Focuses on complex IP issues faced by government contractors, including patent rights, rights in technical data, and computer software, with emphasis on the aerospace, defense, and intelligence sectors. Represents clients in IP disputes and negotiations with the US government. Named one of DC's "Super Lawyers" for Intellectual Property multiple times (2013, 2015–2020, 2023). Frequent speaker and author on IP, and co-instructor of the Federal Publications Seminars Government Contract Intellectual Property Workshop and the Public Contracting Institute's Demystifying Intellectual Property in Government Contracting series. • **Brady Forest** - Founder of Ignite Talks, a global talk series. oard Member, x-Head of Business Development at Sentz Global.Board Member at Ideas Beyond Bordersand Community Arts Stabilization Trust Guest Host and Curator at TED Conferences. Ex - Venture Partner, GenLab Venture Studio

	• **Dr. Yoo Taek Lee** - Executive Dean of International Schools (Solbridge, Endicott & JCFS) and Dean of Endicott College & JWKIM College of Future Studies. For more than 20 years he worked at Boston University as an Operation Professor (recognized as the best professor by student surveys). Author of a few books in Operation Management. Helped in building operation Management in multinational corporations such as GAP, Samsung and others. • **Joleen Liang** - Co-Founder & International CEO, Squirrel Ai Learning, a Chinese Unicorn, with over 60,000+ full-time primary and secondary schools. Squirrel Ai Learning's smart education initiatives earned the 'AI Education Innovation Award' from UNESCO in 2020 and recognition in the inaugural edition of TIME Magazine's World's Top EdTech Companies 2024. China's Power Business Women by Forbes in 2025. Joleen also holds position as an Executive director of the AI and Robotics Education Committee of the China Education Development Strategy Society • **Eva Coates** - is a Consultant Paediatric Surgeon with 20 years of National Healthcare Service in the UK and one of the UK's leading voices on Clinical AI education. Founder of Clinical AI Partners Oxford, she works with NHS Trusts and global health partners to bridge the AI fluency gap - helping clinicians work safely and effectively alongside AI tools. Her platform, Lyra, spans 11 product lines from hackathons and micro-learning to workforce audit tools and regulatory training. • **Pradeep Menon** - Serial entrepreneur in Cybersecurity and AI. Co-Founder and CEO at InceptAI. Chief AI and Information Security Officer (CAISO) at Paramount Computer Systems. Member oa AI for developing countries. Director at Cybersecurity-NxxT. Adjunct Professor of Practice at School of Artificial Intelligence of Amrita Vishwa Vidyapeetham • **Gerald Perez** - CEO of Interactive Brokers, widely recognized as one of the most advanced and respected brokerage firms in the global market, with a market capitalization of around 100 billion dollars • **Joe Ward** - Account Director at Snowflake. One of the world's leading data and AI platform companies, which helps organizations bring together their data, systems, and teams so they can use data and AI in a much more connected and effective way. Snowflake's market capitalization is about $57.49 billion, Warren Baffet invested around Snowflake's IPO in 2020 and Reuters described Snowflake's debut as the largest U.S. listing of that year. • **Chi Chiu** - Founder of CosmicBrain AI. (Building smarter robots for the future with AI), Ex serial Intrapreneur & Technical Leader with Global Operations Expertise at Google, Inc. with over 25 years of experience building successful ventures worldwide and leading multiple Google products from incubation to scaling, working directly with Sergey Brin and Larry Page • **Mahi de Silva** - Co-Founder and Chief Strategy Officer at Higgsfield. Ex executive Vice President, Consumer Mobile Opera Software ASA. Ex Chief Executive Officer Triller. Board Member and Advisor in several startups. Silicon Valley veteran with deep experience building, scaling, and commercializing technology companies. • **James D. Boyd** - Chief Information Officer and Vice President, Digital Technology & Solutions at Valeura Energy (one of largest energy company with HQ in Canada) in the Asia-Pacific region. Principal Consultant, Global CIO Advisor in Digital Transformation, AI & Cyber Infrastructure

	• **Luca Albertalli** - Senior Director of Product Management at Guild, an unicorn, has been named on Fortunes ChangetheWorld list. Co-Founded the Theneeds, Inc. Acquired by shopkick, Inc. • **Irakli Chikava - Co-founder and CEO of Telagri -** an agri-fintech platform helping banks make smarter agricultural lending decisions through real-time farm performance and risk insights. Based in Tbilisi, Georgia, operating across Eastern Europe and Central Asia Startup 15+ years in agriculture and agribusiness consulting, with involvement in $40M+ of investment projects across emerging markets. World Cup Finalist, Speaker, Agritech Professional. Co-organizer of Future Agro Challenge Georgia. Co-Founder and Commercial Director at Agro Solutions Group
Local partners: 6	• **Bolat Sultankulov** - Founder, CEO at ARLAN BIOTECH. StartX S24 cohort Co-Founder, CEO at Astana Biomed Group. Founder, CEO at Kletochnaya terapiya. Doctor of Philosophy - PhD, Bioengineering and Biomedical Engineering at Nazarbayev University. • **Talgat Kantayev** - Co-founder of InvestBanq (Singapore). , Advisor at Bueno.money With more than 8 years in VC (Paladigm Capital,Eurasia RED) • **Sanzhar Rakhmetzhanov** - Co-Founder & Chief Engineering Intelligence Officer at Armeta, Alchemist graduate, a winner at the Energy Venture Day&Pitch Competition by S&P 500 • **Aset Abdualiev** - Founder & CEO at Silkroad Innovation Hub. Founder of Stanford Central Asia Hub Startup Mentor at Alchemist Accelerator Founding Partner at BuildTech VC Board Trustee at American University of Central Asia (AUCA) • **Magzhan Madiyev** - Founding CEO of Astana Hub. Co-Founder of Naimi.kz. Member of the Board of Directors, Independent Director at Astana Innovations, Republican Center for Space Communication, Member of the Board of Directors at Grand Mobile, Axellero.io • **Askar Aituov** - Founder of Devs.kz. Ex-Google Community Manager. Senior Lecturer and a PhD Candidate in Information Systems at Kazakh-British Technical University. Managed the Silkway Accelerator (Google for Startups). Ex Developer and Startup relations for Google developers ecosystem (via EPAM)

2. Goals, objectives and learning outcomes of the program

Program goals	The goal of AI-SANA Program 2 is to equip participants with advanced applied AI skills, operational expertise, and the entrepreneurial capabilities needed to design and implement AI-powered solutions to real industry challenges in Kazakhstan and globally. Where the foundational AI and entrepreneurship program equipped students to understand, analyze, and ideate around AI, Program 2 challenges them to build, lead, and execute. Students develop deep learning proficiency, industry knowledge intelligence, hard and soft entrepreneurial skills, and an industry-embedded capstone that requires genuine solution design - not theoretical proposals. Program 2 graduates are prepared to take leadership roles in AI-enabled organizations, found and scale AI ventures, and contribute meaningfully to Kazakhstan's National AI Strategy.
Program objectives	Upon completing this program, participants will: • Master deep learning architectures including neural networks, CNNs, transformers, and large language models, and apply them to Kazakhstan's priority industries.

	• Prototype AI solutions using professional tools including Google Colab, Hugging Face, and open-source AI frameworks for image recognition, NLP, and recommender systems. • Analyze organizational industry knowledge to identify where AI creates genuine, measurable value - and where it does not. • Apply financial modeling, unit economics, customer discovery, and go-to-market strategy to AI ventures from product concept to early traction. • Develop and exercise the Six C's of entrepreneurial leadership: Communication, Collaboration, Creativity, Critical Thinking, Compassion, and Commitment. • Evaluate AI opportunities across 10+ industry verticals through practitioner-led case studies spanning finance, energy, mining, construction, agriculture, robotics, healthcare, edtech, and AI infrastructure. • Design a credible, team-built AI solution to a specific industry challenge, defended through structured inquiry across all stages of venture development.
Learning outcomes	After successful completion of the program, participants will be able to: • LO1: Apply Python programming and AI fundamentals to develop and customize pre-trained AI models for real-world tasks including image recognition, natural language processing, and recommender systems. • LO2: Prototype and deploy AI applications using professional development environments including Google Colab and pre-trained open-source models, applying customization techniques to industry-specific tasks. • LO3: Analyze organizational processes and supply chains using quantitative and qualitative tools, identify AI-tractable improvement opportunities, and evaluate trade-offs between responsiveness and efficiency. • LO4: Apply customer discovery, MVP development, unit economics, and go-to-market strategy to an AI product, using frameworks drawn from practitioners who have built, funded, and scaled AI companies. • LO5: Demonstrate the Six C's of entrepreneurial leadership (Communication, Collaboration, Creativity, Critical Thinking, Compassion, Commitment) in team settings, including authentic storytelling, feedback delivery, and stakeholder communication. • LO6: Evaluate AI industry opportunities across 10+ sectors using structured frameworks including the AI infrastructure stack, operational automation taxonomy, and industry value chain analysis in targeted industries. • LO7: Design and defend a team-built AI solution to an industry challenge, generating structured inquiry questions through the SMILE platform across all stages of solution development. • LO8: Assess the legal, financial, and ethical dimensions of AI ventures, including IP strategy, funding mechanics, and responsible deployment governance.

3. Course description

Brief description	AI-SANA Program 2, "Digital Solutions to Industry Challenges through AI and Team Leadership," is a 16-week advanced online program for students who have completed the foundational AI and entrepreneurship curriculum. Developed in collaboration with Stanford University's AIRE Program, Zemex Inc., and a network of 34 international practitioners, and nd 6 local practitioners with a track record of results on the global stage, Program 2 takes students from AI fluency to

	AI application - building the technical depth, operational intelligence, and entrepreneurial execution skills needed to create real solutions to real problems. The program is structured as six interconnected modules delivered over 16 weeks. Module 2.1 develops students' technical AI skills through hands-on programming with pre-trained open-source models - covering Python basics and advanced techniques, AI and machine learning fundamentals, image recognition, natural language processing, recommender systems, and model customization. Module 2.2 develops students' industry and operational analysis skills through foundational operations and supply chain frameworks combined with practitioner-led case studies from experts who have deployed successful AI solutions - covering process flow analysis, cycle time and bottlenecks, capacity planning, inventory management, supply chain structure, and AI applications across education, construction in energy, robotics and operational automation, healthcare, mining, minerals & robotics, fintech, cybersecurity, energy, talent management, media/marketing, telecom/data centers/computing infrastructure, and food & agriculture. Module 2.3 develops students' market analysis and venture thinking through trend sessions led by VC investors and ecosystem builders - covering AI opportunities and market maps across healthcare, fintech, cybersecurity, mining, education, operational automation, computing infrastructure, robotics and construction, energy, and entertainment and media, alongside Central Asia-to-global pathways through entrepreneur and investor networks such as Astana Hub, Silk Road Innovation Hub, and developer bridges to Silicon Valley. Module 2.4. builds the commercial capacity to turn AI prototypes into viable businesses, anchored in a problem-first framework of six filtering criteria that distinguish genuine market pull from founder push - priority, quantifiable ROI, speed to value, end-user adoption, dissatisfaction with the status quo, and AI feasibility - covering customer discovery, evaluation design, prototype building, design partner recruitment, market sizing and beachhead segmentation, competitive defensibility, AI unit economics and pricing, and founder-led sales through to a scaled go-to-market function, alongside a "commercial moat" lens that treats workflow integration, niche focus, and switching costs as the durable defensibility in an era where foundation models commoditize technical advantage overnight. Module 2.5 builds the soft skills that underpin entrepreneurial leadership through Stanford GSE–designed sessions anchored in Paul Kim's Six C's framework - Communication, Collaboration, Creativity, Critical Thinking, Compassion, and Commitment - covering cross-cultural communication, persuasion and storytelling, negotiation and public speaking, team building and conflict resolution, servant and ethical leadership, emotional intelligence, decision-making under uncertainty, and personal resilience, alongside an "After Generative AI" lens that treats empathy, cultural fluency, and judgment as the founder's competitive edge where AI produces content but humans produce meaning. Module 2.6 is the capstone final project, where teams design and pitch a Minimum Viable Product (MVP) or venture product/service that leverages AI skills learned throughout the program - presented as a Google Slides deck and YouTube pitch video to simulate a real investor or industry pitch. Assessment throughout uses the Stanford SMILE platform, integrating knowledge exams, inquiry assessments, case analyses, open activities, and hands-on experiments. All video content is produced and delivered via private YouTube with LMS integration through Platonus.

4. Teaching methods

Teaching methods	Program 2 employs a blended applied-learning approach. Where the foundational program built understanding, this program builds execution. All teaching methods are designed to develop students' ability to apply what they learn to real problems in real organizations. • Video lectures (required and optional) - 155 video segments (74,94 hours) delivered across six production formats. Required videos cover core content; optional segments provide deeper exploration. Expert practitioner lectures by Bola Adegbulu, Dr. Yoo Taek Lee, Dr. Paul Kim, and 37 industry guests. • Reading and research materials - Comprehensive reading materials for each module, case studies, and technical references. Estimated reading time 12 hours per module. • Hands-on experiments with AI tools - Google Colab labs for working with pre-trained AI models, prototyping AI applications, image recognition, NLP pipelines, and recommender system experiments. In preparing their capstone pitch, students engage with the underlying analytical work any credible venture case must reflect: process mapping, supply chain feasibility reasoning, customer discovery, and financial modeling, as integrated reasoning within the pitch rather than a separate assigned artifacts. • SMILE-based inquiry learning - The Stanford SMILE platform is used for student-generated inquiry questions, industry case analyses, knowledge exams, and the structured capstone inquiry. All assessment uses Bloom's Taxonomy Level 4+ and AI-evaluated scoring. • Team interaction and collaborative inquiry - Weekly team sessions for peer discussion and collaborative problem-solving. Six C's exercises throughout Module 2.5. • Project-based learning - The capstone project requires teams to select a real industry challenge, define an AI-powered solution, and pitch a Minimum Viable Product (MVP) that leverages AI skills learned throughout the program - presented as a Google Slides deck and YouTube pitch video, using the SMILE platform. • Expert-led content - Insights from founders, venture capitalists, CFOs, legal practitioners, and AI engineers who provide real-world perspectives on building and scaling AI solutions.

5. Course policy

Course policy	**Participation Policy:** Students are expected to dedicate approximately 15–17.5 hours per week to the program. This includes required video lectures, reading materials, hands-on experiments, team interactions, and SMILE assessments. Active participation in all program components is required for successful completion. **Completion and Grading Policy:** This program operates on a Pass / No Pass basis. The passing threshold is 70% on all SMILE activities. Students who complete all SMILE activities with a passing result will receive the AI-SANA Program 2 Certificate of Completion. **Deadline Policy:** All SMILE assessments, hands-on experiments, and weekly assignments must be submitted by the deadlines specified in the course schedule. Submissions are made through the Platonus LMS and SMILE platforms. Late submissions may receive reduced credit at the discretion of program coordinators. **Weekly Commitment Breakdown:**

	• Video lectures (required + optional): 4.5 hours/week (2 hrs required + 2.5 hrs optional) • Reading and research: 4 hours/week • Hands-on experiments: 4 hours/week • Team project discussion: 1 hour/week • Exam assignment: 4 hours/week **Communication Policy:** All course-related inquiries must be addressed according to Operational Scenario. Communication should remain professional, respectful, and relevant to the subject matter. **Inclusion and Individual Needs Policy:** This program is committed to inclusive practices. The program coordinators will take into account participants' individual learning needs when organizing the learning process. Participants are encouraged to communicate any special requirements in advance.

6. Academic integrity

Academic integrity	**Plagiarism and other violations policy:** Plagiarism, fabrication of results, falsification of data, or the use of unauthorized materials during assignments and project work are strictly prohibited. Repeated violations may lead to removal from the program. **AI Tools Usage Policy:** Given the nature of this program, the use of AI tools (e.g., ChatGPT, Claude AI, Gemini, Copilot) is permitted and encouraged under the following conditions: • AI may be used for brainstorming, code debugging, drafting documentation, and accelerating research, but must not replace independent analysis and critical judgment. • All project deliverables must reflect the participant's own thinking and domain understanding. • If AI tools are used in submissions, participants must clearly indicate which parts were AI-assisted. • Submissions entirely generated by AI without participant evaluation or original contribution constitute academic misconduct. • Active use of GenAI as a learning accelerator is highly recommended throughout all modules. **Capstone Project** (The final stage) **- Important:** For the capstone project (Module 2.6), teams must NOT use AI to generate their pitch content on their behalf. The pitch video and slide deck must reflect the team's own thinking, research, and original venture concept. Submissions that are entirely AI-generated without genuine team contribution will be considered academic misconduct. **Code of Academic Integrity:** 1. Honesty - All assignments, experiments, and project documentation must be the participant's own work. 2. Fairness - Participants must respect deadlines, assessment rules, and the evaluation system. 3. Respect - Participants should engage in academic discussions respectfully, valuing diverse perspectives.

7. Literature and resources

A. Required program materials

1. AI-SANA Module 2.1 Reading Material: Deep Learning Foundations & Applications. AI-SANA Program, 2025. (12 hours) [M2.1]
2. AI-SANA Module 2.2 Reading Material: Why Operations Matter in the Era of AI. AI-SANA Program, 2025. (12 hours) [M2.2]
3. AI-SANA Module 2.3 Reading Material: Trends and Case Studies in Local and International Markets. AI-SANA Program, 2025. (12 hours) [M2.3]
4. AI-SANA Module 2.4 Reading Material: Hard Entrepreneurial Skills. AI-SANA Program, 2025. (12 hours) [M2.4]
5. AI-SANA Module 2.5 Reading Material: Soft Skills for Leadership and Teamwork. AI-SANA Program, 2025. (12 hours) [M2.5]

B. Supplementary literature

Deep Learning and AI Architecture:

1. Goodfellow, I., Bengio, Y. & Courville, A. Deep Learning. MIT Press, 2016. [M2.1]
2. Nielsen, M. Neural Networks and Deep Learning. Online, 2019. [M2.1]
3. Géron, A. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow (3rd ed.). O'Reilly, 2022. [M2.1]
4. Vaswani, A. et al. "Attention Is All You Need." NeurIPS, 2017. [M2.1]

Industry Knowledge Management:

1. Eliyahu M. Goldratt and Jeff Cox, The Goal: A Process of Ongoing Improvement. [M2.2]
2. Hau L. Lee, V. Padmanabhan, and Seungjin Whang, 'The Bullwhip Effect in Supply Chains,' Sloan Management Review, 1997. [M2.2]
3. Marshall L. Fisher, 'What Is the Right Supply Chain for Your Product?', Harvard Business Review, 1997. [M2.2]

Entrepreneurship and Venture Building:

1. Ries, E. The Lean Startup. Crown Business, 2011. [M2.4]
2. Fitzpatrick, R. The Mom Test. Self-published, 2013. [M2.4]
3. Feld, B. and Mendelson J. Venture Deals. Wiley, 2019. [M2.4]
4. Thiel, P. Zero to One. Crown Business, 2014. [M2.4]
5. Horowitz, B. The Hard Thing About Hard Things. HarperBusiness, 2014. [M2.4]

Leadership and Human Skills:

1. Lencioni, P. The Five Dysfunctions of a Team. Jossey-Bass, 2002. [M2.5]
2. Scott, K. Radical Candor. St. Martin's Press, 2017. [M2.5]
3. Kahneman, D. Thinking, Fast and Slow. Farrar, Straus and Giroux, 2011. [M2.5]
4. Voss, C. Never Split the Difference. HarperBusiness, 2016. [M2.5]
5. Dweck, C. Mindset: The New Psychology of Success. Ballantine Books, 2007. [M2.5]
6. Mollick, E. Co-Intelligence: Living and Working with AI. Portfolio, 2024. [M2.5]

C. Online resources and tools

AI Tools and Platforms:

1. Google Colab - https://colab.research.google.com/ [M2.1]
2. Hugging Face - https://huggingface.co [M2.1]
3. Gradio - https://gradio.app [M2.1]
4. Stanford SMILE Platform [All modules]

Industry Research and Reports:

1. Stanford HAI Annual AI Index - https://aiindex.stanford.edu [M2.5]
2. The Future of Food and Farming (UK Government Foresight Report) [M2.3]

3. McKinsey Global Institute: The Age of AI - The macro-level analysis of AI's economic impact across industries [M2.3]
4. MIT Center for Transportation and Logistics research publications (ctl.mit.edu) [M2.2]
5. Astana Hub Kazakhstan research and program documentation - https://astanahub.com [M2.3]
6. World Bank Digital Development in Central Asia Reports [M2.1]

Entrepreneurship Resources:

1. Astana Hub research and program documentation. Current programs, resident stories, and ecosystem data for Kazakhstan's primary tech hub - astanahub.com [M2.3]
2. a16z Essays and reports on technology business strategy and AI market dynamics - https://a16z.com [M2.4]
3. Silk Road Innovation Hub podcast and content. English-language content specifically for Central Eurasian founders navigating Silicon Valley - silkroadhub.io [M2.3]
4. The Startup Community Way by Brad Feld and Ian Hathaway - How to build and participate in a thriving innovation ecosystem [M2.3]

Kazakhstan and Central Asia:

1. Digital Kazakhstan State Program [M2.1]
2. AIFC Research on Financial Technology in Kazakhstan - https://aifc.kz [M2.5, M2.4]
3. World Bank Digital Development in Central Asia Reports [M2.5, M2.4]
4. Astana Hub Reports on Kazakhstan's Tech Ecosystem - https://astanahub.com [M2.5, M2.4]

8. Forms of control and assessment

All assessments are conducted through the SMILE platform. The program operates on a Pass / No Pass basis with a 70% threshold on all SMILE activities.

No.	Type of Work	Form of Control	Evaluation Criteria
1	SMILE Knowledge Exam	Exam assignment. Online assessment (Every module)	Correctness and completeness of answers; mastery of key concepts and module content. Pass: ≥70%
2	SMILE Inquiry Assessment	Discussion of a team project. Student-generated questions (Every module with team)	Quality and depth of questions; critical thinking demonstrated; Bloom's Taxonomy Level 4+ and AI evaluation score 5+/10. Pass: ≥70%
3	SMILE Case Analysis	Practical experiment. Case study analysis (Every Module)	Ability to analyze industry cases, identify failure modes, propose AI-powered solutions, and apply learned frameworks. Pass: ≥70%
4	Open SMILE Activities	Discussion of a team project Creativity exercises (Selected modules)	Ability to connect concepts across domains; creativity, critical thinking, and content mastery. Pass: ≥70%
5	Hands-on Experiments	Coding Handbook. Lab reports and code outputs from Coding Handbook (12h) (Fundamentals of	Correct implementation of deep learning models; functional code and deployable applications; quality of analysis. Pass: ≥70%

		Artificial Intelligence Programming)	
6	Final Capstone Project	Google Slides MVP pitch deck + YouTube pitch video (Module 2.6)	Evaluated by AI across 12 weighted pitch stages (120 pts total, normalized to 100). Stage 3 Solution/Product carries highest weight (15 pts). Without slide deck, 20% score penalty applies. Resubmission allowed. Pass: ≥70/100

9. Grading system

This program uses a Pass / No Pass grading system. There is no letter grading or GPA calculation.

Result	Threshold	Criteria
PASS	≥ 70% on all SMILE activities	The participant has completed all required SMILE activities (Knowledge Exams, Inquiry Assessments, Case Analyses, Open SMILE Activities, Hands-on Experiments) with a score of 70% or above on each, and has successfully completed the Capstone Project with the required quality standards. The participant receives the AI-SANA Program 2 Certificate of Completion.
NO PASS	< 70% on any SMILE activity	The participant has not met the 70% threshold on one or more SMILE activities, or has not completed the required Capstone Project. The participant does not receive the Certificate of Completion.

Note: There is no intermediate grading between Pass and No Pass. The 70% threshold applies to each individual SMILE activity.

10. Course schedule

Week	Module & Key Subtopics	Video (Req)	Video (Opt)	Reading & Research	Hands-on Practical experiment	Team Discussion of a team project.	Assess. Exam assignment
1–3	Module 2.1. Fundamentals of Artificial Intelligence Programming - Hands-on Python in Google Colab s - Progression from first principles to production-grade AI systems Neural Network Foundations - Build a neuron from scratch and understand why linear models fail on nonlinear problems (the “fox problem”) - Develop intuition for backpropagation and gradient-based learning - Train a first deep learning model on MNIST using TensorFlow and Keras Computer Vision with CNNs: - Why images need specialized architectures: convolutional filters, pooling, dropout, batch normalization, and data augmentation - Build a first CNN from the ground up - Apply transfer learning with MobileNetV2, with ResNet, DenseNet, and EfficientNet as alternatives - Medical imaging segment: deep learning vs. traditional ML (SVM with handcrafted features) to teach when each approach wins Language & Transformers	6h	7,97h	12h	12h	3h	12h

	- Word embeddings and the geometry of meaning in vector space - Self-attention, multi-head attention, positional encoding, and the BERT / GPT / T5 families - Build a story generator and fine-tune GPT-2 on Shakespeare-style text using Hugging Face - How modern LLMs (ChatGPT, Claude, Gemini) work under the hood - Advanced prompt engineering: zero-shot, few-shot, and chain-of-thought - LLM API integration, a Kazakh question-answering assistant, and shareable web apps via Gradio Customization - Fine-Tuning & RAG - Fine-tune with QLoRA and evaluate results across multiple dimensions - RAG pipeline with sentence-transformers (all-MiniLM-L6-v2) and FAISS, plus interactive embedding-galaxy visualization - Responsible AI: bias, hallucination, and trust - Capstone: conversational home robot combining intent detection (embeddings), RAG memory (FAISS), templated response generation, and Q-learning for navigation - four techniques working together in one system Applied AI & Entrepreneurship - Three-question framework for choosing between prompting, RAG, fine-tuning, CNNs, traditional ML, and reinforcement learning - Approach mapping onto eight AI SANA priority industries (mining, agriculture, healthcare, education, fintech, construction, telecom, energy) - The real cost of AI in production: infrastructure and compliance typically dwarf API spend - Pitch toolkit and “your AI journey” reflective synthesis preparing students to propose their own ventures						
4–6	Module 2.2. Industry Knowledge - Process fundamentals: process flow diagrams, SIPOC, and levels of process - why operational understanding must precede AI adoption - Process analysis: task time, throughput time, cycle time, capacity, bottlenecks, and line balancing as the lens for locating genuine AI ROI - Levers for improvement: available time, resources, and task time - their trade-offs and how AI reshapes each in the shift toward smart factories - Supply chain basics: global flows, lead time, responsive vs. cost-efficient designs, and the bullwhip effect - framing AI's role in forecasting and coordination - Inventory and industry-specific supply chains: EOQ, inventory functions, and push/pull and buyer/producer-driven structures across food, construction, healthcare, energy, and mining - where industrial AI fits Industrial case studies: - Education - Healthcare - Cyber security in Telecom - Fintech - Construction and robotics - Construction in Energy sector - Data management and cloud computing	6h	10,48h	12h	12h	3h	12h

	- Robots in Mining - Content creation for Social Media marketers - Energy - Talent Management - Agriculture						
7–9	Module 2.3. Trends and Case Studies in Local and International Markets - How global venture capital reads markets: sector selection, market sizing, and where investable white spaces appear -Market mapping as a transferable analytical tool across unfamiliar industries - Where AI creates durable value versus where it remains a thin wrapper - Domain expertise and data access as defensive moats against foundation-model commoditization - Timing, constraints, and founder conviction as real factors in venture outcomes - Pathways from Central Asia to global markets: programs, networks, and the concrete mechanics of crossing borders Market mapping of industries: - healthcare & life sciences - fintech - decentralized finance (DFAI) - cybersecurity - minerals & mining - education & edtech - operational automation - computing infrastructure - robotics - energy - food & agriculture - entertainment, media & creative industries.	6h	8,59h	12h	12h	3h	12h
10-12	Module 2.4. Hard Entrepreneurial Skills - Problem discovery and product-market fit (six-criteria filter) -AI evaluation, prototyping, and MVP iteration with design partners -Competitive defensibility and foundation-model platform risk -Customer segmentation and beachhead strategy -AI unit economics, pricing, and financial modeling - Founder-led sales and go-to-market for AI products - Food & Agriculture: Analyzing the Industry, Building Go To Market, and Winning Over VCs - Biotech & Life Sciences: How to Enter Deep Tech, Build a Small AI-Leveraged Team, and Win Over Top AcceleratorsAI Applications: How to Define - - - Your Ideal Customer Profile, Build Moats Against Foundational Players, and Raise Capital Judiciously -AI Products in Cybersecurity: How to Start from the Problem, Build Iteratively, and Defend Against Foundation Models - Sales in B2B Enterprise Tech: How to Sell Before You Have a Product, Run the Full Sales Funnel, and Hire Your First Sales Team - Fintech & Wealth Management: How to Build AI Products in a Trust-Based Industry, Segment High-Net-Wealth Clients, and Scale Cross-Cultural Teams	6h	9,23h	12h	12h	3h	12h

	- Startup Finance in Tech: How to Master Financial - - Statements, Understand CAC-to-LTV, and Balance Growth vs. Profitability - Startup Legal for AI Companies: How to Architect Your Company, Structure for Global VC Investment, and Protect IP in the AI Era - Communication for Founders in the AI Era: How to Know Your Audience, Tell Stories That Land, and Use AI Without Losing Your Voice						
13–15	Module 2.5. Soft Skills for Leadership and Teamwork Six C's framework (Communication, Collaboration, Creativity, Critical Thinking, Compassion, Commitment) Cross-cultural communication and global business fluency Persuasion, negotiation, and public speaking Storytelling and personal branding Networking and relationship building Collaboration, team dynamics, and conflict resolution Leadership development and personal growth Emotional intelligence and servant/ethical leadership Critical thinking, creativity, and decision-making under uncertainty Resilience, growth mindset, and continuous learning	6h	8,09h	12h	12h	3h	12h
16	Module 2.6. The Final Stage • Google Slides pitch deck preparation (12-stage pitch format) • YouTube pitch video recording and upload (with MVP)	0.5h	0h	0h	0h	0h	10h
TOTAL	**16 Weeks (6 modules)**	**30,5 h**	**60,84 h**	**60 h**	**60 h**	**15 h**	**60 h**

Program 2 Total: 235,5 hours compulsory + 60,84 hours optional = 279,86 hours total | 74,86 hours video content (30,5 hours compulsory)

11. List of topics and assignments by module

No.	Module / Topic Title	Hours for assessments (Comp.)	SMILE Assessment Types	Deadline
2.1	Fundamentals of Artificial Intelligence Programming	51	Practical experiment (Case study analysis) Discussion of a team project. (Inquiry. Student-generated questions). Knowledge Exam (Exam assignment) Hands-on Experiments. (Coding Handbook. Laboratory work)	End of Week 3
2.2	Industry Knowledge	39	Practical experiment (Case study analysis) Discussion of a team project. (Inquiry. Student-generated questions). Knowledge Exam (Exam assignment)	End of Week 6
2.3	Trends and Case Studies in Local and International Markets	39	Practical experiment (Case study analysis) Discussion of a team project. (Inquiry. Student-generated questions).	End of Week 9

No.	Module / Topic Title	Hours for assessments (Comp.)	SMILE Assessment Types	Deadline
2.1	Fundamentals of Artificial Intelligence Programming	51	Practical experiment (Case study analysis) Discussion of a team project. (Inquiry. Student-generated questions). Knowledge Exam (Exam assignment) Hands-on Experiments. (Coding Handbook. Laboratory work)	End of Week 3
2.2	Industry Knowledge	39	Practical experiment (Case study analysis) Discussion of a team project. (Inquiry. Student-generated questions). Knowledge Exam (Exam assignment)	End of Week 6
			Knowledge Exam (Exam assignment)	
2.4	Hard Entrepreneurial Skills	39	Practical experiment (Case study analysis) Discussion of a team project. (Inquiry. Student-generated questions). Knowledge Exam (Exam assignment)	End of Week 12
2.5	Soft Skills for Leadership and Teamwork	39	Practical experiment (Case study analysis) Discussion of a team project. (Inquiry. Student-generated questions). Knowledge Exam (Exam assignment)	End of Week 15
2.6	The Final Stage: Capstone Submission (Google Slides pitch deck + YouTube pitch video, including prototype demonstration)	10	Google Slides pitch deck (12-stage MVP pitch format) YouTube pitch video (min. 3 min / 180 sec, English captions) AI-evaluated across 12 weighted stages (120 pts → 100)	End of Week 16

Language of Tests and Exams (SMILE)

All assignments on the **SMILE** platform are initially provided **in English**.

At the same time, the course lessons are available in **three languages: Russian, English, and Kazakh**. If a student is more comfortable working in another language, the student may:

- use the **browser translation function** to view the tasks in a preferred language;
- **copy and translate** both questions and answer options using third-party translation services.

This format is used for two reasons:

1. **Academic reason**
   Working with tasks in English is part of the program's educational approach and helps develop the ability to overcome small practical challenges, look for solutions, and adapt independently to real-world conditions in a modern digital environment.

Technical reason

A significant portion of the tasks and assessment mechanisms is generated using artificial intelligence tools. At the current stage, these tools provide the highest quality generation and evaluation **in English**.

12. Capstone Project: The Final Stage for an Industry Challenge

Overview

The capstone project is the culminating assessment of Program 2. Having covered AI programming fundamentals, industry knowledge, global trends, hard entrepreneurial skills, and leadership development, participants now form teams to design and pitch a Minimum Viable Product (MVP) or venture product/service that leverages AI concepts and tools learned throughout the program (including MVP).

The MVP or venture concept must be grounded in a real industry problem and incorporate AI as a core component of the solution. Teams submit two deliverables: a Google Slides pitch deck and a YouTube pitch video presenting the MVP as if to real investors or industry partners.

Assignment

Teams will spend 10 hours on this capstone project. The submission consists of two deliverables: (1) a Google Slides pitch deck describing the Minimum Viable Product (MVP) or venture product/service that leverages AI learned from the program, and (2) a YouTube pitch video presenting the MVP as if to real investors or industry partners. Before starting, teams should review the sample materials and format guidelines provided in the LMS.

Deliverables

1. Form a team and confirm the MVP or venture product/service concept - it must leverage AI tools and skills learned in the program.

2. Prepare a pitch slide deck using Google Slides (not PowerPoint or Keynote) covering all 12 pitch stages: Problem Definition, Target Market, Solution/Product (MVP), Value Proposition, Business Model, Competitive Analysis, Go-to-Market Strategy, Financial Projections, Team, Traction/Milestones, Funding Ask, and Vision/Impact. Set sharing to "Anyone with the link can view" before submitting.

3. Record a startup venture pitch video presenting the MVP, upload to YouTube, and submit the YouTube URL. The video must have English captions or transcript available and be a minimum of 3 minutes in length (180 seconds).

Important Guidelines

• The slide deck must be Google Slides with link sharing set to "Anyone with the link can view." PowerPoint and Keynote files are not accepted. Without a slide deck, only 80% of the AI score counts toward passing (20% penalty).

• The pitch video must be published to YouTube (youtube.com or youtu.be) with English captions or transcript. It must be a startup venture pitch presentation and a minimum of 3 minutes (180 seconds) in length.

• The pitch video must feature all team members presenting.

• The MVP concept must reflect the team's own thinking and original venture idea. Pitch content that is entirely AI-generated without genuine team contribution will be considered academic misconduct.

• Submissions that do not include both the pitch video URL and the Google Slides URL will receive a reduced score.

Evaluation Criteria

The capstone is evaluated by AI across 12 startup pitch stages with a weighted rubric totaling 120 points, normalized to a 0–100 scale. Formula: overall score = sum of all stage scores ÷ 120 × 100. Pass threshold: 70/100. Resubmission is allowed - students may resubmit until they reach 70/100.

Stage	Topic	Max Score	Notes

1	Problem Definition	10	
2	Target Market	10	
3	Solution / Product (MVP)	15	Highest weight. Sub-criteria: Clarity (4 pts), Demonstration Logic (4 pts), Innovation (4 pts), Meaningfulness (3 pts). Score of 11+/15 requires all four to be clearly present.
4	Value Proposition	10	
5	Business Model	10	
6	Competitive Analysis	10	
7	Go-to-Market Strategy	10	
8	Financial Projections	8	Reduced weight - students have limited real data
9	Team	10	
10	Traction / Milestones	7	Reduced weight - students have limited real data
11	Funding Ask	10	
12	Vision / Impact	10	
	Total	**120**	**Normalized to 100. Pass: ≥ 70/100**

All stages are assessed on: clarity, depth, specificity, evidence of research, feasibility, quality of reasoning, completeness, and conviction.

13. SMILE activity links by module

The following links provide direct access to all SMILE platform activities and reading materials for each module. Students should access these through the Platonus LMS or directly via the links below.

Module 2.1: Fundamentals of Artificial Intelligence Programming

Reading & Research - 12 hours
https://docs.google.com/document/d/1obHG5HTWNxHNeK8qQxf_ej0a64b3JFdrk-8cb27nhfM/edit?usp=sharing

Coding Experiment Handbook -
https://docs.google.com/document/d/1pnSj6ZeSxm3Ud_-TGgotEtSQjQDDfBxj/edit?usp=sharing&ouid=100913751170041539879&rtpof=true&sd=true

Practical experiment 1 -
https://newsmile.seedsofempowerment.org/join-activity?code=71z855OKeXOxKi8WkR1jWQ

Discussion of a Team Project 1 -
https://newsmile.seedsofempowerment.org/join-activity?code=fP2bpC9TnQOco7F-v7xDQg

Exam assignment 1 -
https://newsmile.seedsofempowerment.org/join-activity?code=KMUPCjXLtQeT8fHrxyAePg

Practical experiment 2 -
https://newsmile.seedsofempowerment.org/join-activity?code=9pNxXR0TwMMrWQdASqLuqA

Discussion of a Team Project 2 -
https://newsmile.seedsofempowerment.org/join-activity?code=J3AT8l06NoeQnqSSIP5jCQ

Exam assignment 2 -
https://newsmile.seedsofempowerment.org/join-activity?code=2Q1-mi-UDAg4fS6pqENqHw

Practical experiment 3 -
https://newsmile.seedsofempowerment.org/join-activity?code=0EytUNSriGoC5LZ5lTVp9Q

Discussion of a Team Project 3 -
https://newsmile.seedsofempowerment.org/join-activity?code=oF8_3eOnc6fmGMXuN7gSBg

Exam assignment 3-
https://newsmile.seedsofempowerment.org/join-activity?code=SbJXjty1bfSTL8HN3qDi5A

Module 2.2: Industry Knowledge

Reading and Research - 12 hours -
https://docs.google.com/document/d/1Ecrbn_APlBO9CUTo4ewq8J6dh-aBYNQCCFTeghHCJlk/edit?usp=sharing

Practical experiment 1 -
https://newsmile.seedsofempowerment.org/join-activity?code=gPlNZlfshCRovCIrKp1hCQ

Discussion of a Team Project 1 -
https://newsmile.seedsofempowerment.org/join-activity?code=AWBK3ZnzDZWl7ekE4-HaBg

Exam assignment 1 -
https://newsmile.seedsofempowerment.org/join-activity?code=Kl7C2PhRabpUXseuMQFrkA

Practical experiment 2 -
https://newsmile.seedsofempowerment.org/join-activity?code=fSISSWhU6Ohsl3arxFnO2g

Discussion of a Team Project 2 -
https://newsmile.seedsofempowerment.org/join-activity?code=xwu5I138kF6KhgWGHodKSQ

Exam assignment 2 -
https://newsmile.seedsofempowerment.org/join-activity?code=0KVq7iWm78QPwU9mdBu5KQ

Practical experiment 3 -
https://newsmile.seedsofempowerment.org/join-activity?code=d20cFqMypMmqhEBETuNfdQ

Exam assignment 3 -
https://newsmile.seedsofempowerment.org/join-activity?code=xXo1mMyADWP9ZOcyXjkNAw
Discussion of a Team Project 3 -
https://newsmile.seedsofempowerment.org/join-activity?code=MlzaUjsLAKMUYxYDiANmfQ

Module 2.3: Trends and Case Studies in Local and International Markets

Reading and Research - 12 hours -
https://docs.google.com/document/d/177anGy2b33Lkz3D-ifWe8jr0m7lVS80_DPj_uST8OLY/edit?usp=sharing

Practical experiment 1 -
https://newsmile.seedsofempowerment.org/join-activity?code=SGigwoXJIMNwSLrrYZj_og

Discussion of a Team Project 1 -
https://newsmile.seedsofempowerment.org/join-activity?code=QwRsOzck2Ki1dyiwQtVxbw

Exam assignment 1 -
https://newsmile.seedsofempowerment.org/join-activity?code=AaS15GYwsXtcFdNFnmSmVQ

Practical experiment 2 -
https://newsmile.seedsofempowerment.org/join-activity?code=13TuyI4-zBKS93AAgL7BbQ

Discussion of a Team Project 2 -
https://newsmile.seedsofempowerment.org/join-activity?code=JheVSmTkc3VroUXoU6R-_A

Exam assignment 2 -
https://newsmile.seedsofempowerment.org/join-activity?code=F5Z5HYADD80IxMIa9ujpPw

Practical experiment 3 -
https://newsmile.seedsofempowerment.org/join-activity?code=0VN6qBl0EafwcYl11g6VGQ

Discussion of a Team Project 3 -
https://newsmile.seedsofempowerment.org/join-activity?code=eaKFuF5BNwwn_w0mR8IgrQ

Exam assignment 3 -
https://newsmile.seedsofempowerment.org/join-activity?code=Tl7DxuIIos-HoFjiMLUr2g

Module 2.4: Hard Entrepreneurial Skills

Reading and Research - 12 hours -
https://docs.google.com/document/d/1ck_Drd00RGWTVkYKvsg-N0LJ1zq-b65dPcvNRg5mkSg/edit?usp=sharing

Practical experiment 1 -
https://newsmile.seedsofempowerment.org/join-activity?code=kdTVleeAPMY6vt_XNqN0Yw

Discussion of a Team Project 1 -
https://newsmile.seedsofempowerment.org/join-activity?code=c9gSCo_MxJGN6p7fFvyfkQ

Exam assignment 1 -
https://newsmile.seedsofempowerment.org/join-activity?code=02feEneD7TQJ1RfSI3Yp-A

Practical experiment 2 -
https://newsmile.seedsofempowerment.org/join-activity?code=4bPNPtzw3XG3TQZRTvHXYw

Discussion of a Team Project 2 -
https://newsmile.seedsofempowerment.org/join-activity?code=J1zkprZd6BKk-NUlAuyWLA

Exam assignment 2 -
https://newsmile.seedsofempowerment.org/join-activity?code=qHQveI_sBpB4X3Djfcny-Q

Practical experiment 3 -
https://newsmile.seedsofempowerment.org/join-activity?code=OQkrayHSvFQ2AkmVqms4hg

Discussion of a Team Project 3 -
https://newsmile.seedsofempowerment.org/join-activity?code=Z36Ub-esvJbU398nVUPdgQ

Exam assignment 3 -
https://newsmile.seedsofempowerment.org/join-activity?code=qw4UU5FVTv0SpM8EKW_4Dw

Module 2.5: Soft Skills for Leadership and Teamwork

Reading and Research - 12 hours -
https://docs.google.com/document/d/1hdc0wk7whngzGnRUBmtYEh7jxYC8oDCG6pWtnRZ4iXA/edit?usp=sharing

Practical experiment 1 -
https://newsmile.seedsofempowerment.org/join-activity?code=UxmxkN26u1FmlScUthRQ2A

Discussion of a Team Project 1 -
https://newsmile.seedsofempowerment.org/join-activity?code=JTn1hLScZrSbnOQIwxD8uw

Exam assignment 1 -
https://newsmile.seedsofempowerment.org/join-activity?code=tBI_s4vS6YnP-4yyZ678lA

Practical experiment 2 -
https://newsmile.seedsofempowerment.org/join-activity?code=fUHFQZD63z1ZtH8IPzrxGg

Discussion of a Team Project 2 -
https://newsmile.seedsofempowerment.org/join-activity?code=j9b4wLX_ZlU0LrqUT7rQLw

Exam assignment 2 -
https://newsmile.seedsofempowerment.org/join-activity?code=MpqwCuOi5O8C1vH290ktvQ

Practical experiment 3 -
https://newsmile.seedsofempowerment.org/join-activity?code=V5_LP45F_IZW5dTX_g-Pkg

Discussion of a Team Project 3 -
https://newsmile.seedsofempowerment.org/join-activity?code=px24GAdS6J0yFoSsG7iajQ

Exam assignment 3 -
https://newsmile.seedsofempowerment.org/join-activity?code=cmBkU7EeA7qM8ntHRVjE6w

Module 2.6: The Final Stage

Capstone Submission

https://newsmile.seedsofempowerment.org/join-group?code=o9KDk6MTnKsgSJIpuefMfw

11. List of content

Application 1
AI-SANA NATIONAL AI TRAINING INITIATIVE

№	Speaker	Position	Topic Title	Timing	Req/Opt
Module 2.1. Fundamentals of Artificial Intelligence Programming					
1	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	What Is a Neural Network, Really?	10,11	Required
2	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Building a Neuron from Scratch	43,45	Required
3	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	The Fox Problem - Why One Layer Isn't Enough	9,06	Required
4	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile	Solving the Fox Problem	43,39	Required

		Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank			
5	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	How Neural Networks Learn - Backpropagation	10,41	Required
6	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Training Your First Real Neural Network	41,47	Required
7	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Why Images Need Special Networks	13,56	Required
8	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Understanding Convolutions	32,27	Required
9	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	The CNN Revolution & Data Augmentation	12,07	Required
10	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Building Your First CNN	23,19	Required

11	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Transfer Learning	12,08	Required
12	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Transfer Learning for Medical Imaging	25,22	Required
13	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Words as Numbers - Embeddings	10,05	Required
14	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Exploring Word Embeddings	31,03	Required
15	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	The Transformer Revolution	10,38	Required
16	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Build a Story Generator	31,58	Required
17	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean	Fine-Tuning Transformers	8,53	Required

		and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank			
18	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	See Fine-Tuning in Action	22,32	Optional
19	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	How Modern LLMs Work	9,36	Required
20	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Advanced Prompt Engineering	51,14	Optional
21	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	The API Economy	10,31	Optional
22	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Building a Kazakh Culture Q&A Assistant	27,14	Optional
23	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Creating Interfaces with Gradio	9,23	Optional
24	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco	Building Your First Gradio App	35,25	Optional

		described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank			
25	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Introduction to Fine-Tuning	10,14	Optional
26	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Fine-Tuning with QLoRA	34,22	Optional
27	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Evaluating Fine-Tuned Models + Introduction to RAG	5,44	Optional
28	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Evaluation Framework + Building RAG System	27,4	Optional
29	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	AI Agents: From Chatbots to Autonomous Systems	23,55	Optional
30	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Responsible AI - Bias, Hallucination, and Trust	4,47	Optional

31	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Building a Conversational Home Robot	44,54	Optional
32	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	AI Tool Selection	13,07	Optional
33	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Tool Selection & Responsible AI	30,15	Optional
34	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	From Prototype to Production	5,02	Optional
35	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	AI Production Costs + Startup Ideation	57,03	Optional
36	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	AI Pitch Toolkit	45,59	Optional
37	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean	Your AI Journey	5,11	Optional

		and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank			
Subtotal Module 2.1 Fundamentals of Artificial Intelligence Programming (minutes)				**838,33**	
Module 2.2: Industry Knowledge					
1	Dr. Yoo Taek Lee	Executive Dean of International Schools (Solbridge, Endicott & JCFS) and Dean of Endicott College & JWKIM College of Future Studies. For more than 20 years he worked at Boston University as an Operation Professor (recognized as the best professor by student surveys). Author of a few books in Operation Management. Helped in building operation Management in multinational corporations such as GAP, Samsung and others.	Introduction: Why operations are matter?	2,5	Required
2	Dr. Yoo Taek Lee	Executive Dean of International Schools (Solbridge, Endicott & JCFS) and Dean of Endicott College & JWKIM College of Future Studies. For more than 20 years he worked at Boston University as an Operation Professor (recognized as the best professor by student surveys). Author of a few books in Operation Management. Helped in building operation Management in multinational corporations such as GAP, Samsung and others.	Why Operations matter in the Era of AI?	9,24	Required
3	Dr. Yoo Taek Lee	Executive Dean of International Schools (Solbridge, Endicott & JCFS) and Dean of Endicott College & JWKIM College of Future Studies. For more than 20 years he worked at Boston University as an Operation Professor (recognized as the best professor by student surveys). Author of a few books in Operation Management. Helped in building operation Management in multinational corporations such as GAP, Samsung and others.	Basics of Process	8,53	Required
4	Dr. Yoo Taek Lee	Executive Dean of International Schools (Solbridge, Endicott & JCFS) and Dean of Endicott College & JWKIM College of Future Studies. For more than 20 years he worked at Boston University as an Operation Professor (recognized as the best professor by student surveys). Author of a few books in Operation Management. Helped in building operation Management in multinational corporations such as GAP, Samsung and others.	Process Analysis	17,16	Required
5	Dr. Yoo Taek Lee	Executive Dean of International Schools (Solbridge, Endicott & JCFS) and Dean of Endicott College & JWKIM College of Future Studies. For more than 20 years he worked at Boston University as an Operation Professor (recognized as the best professor by student surveys). Author of a few books in Operation Management. Helped in building operation Management in multinational corporations such as GAP, Samsung and others.	Improvement Opportunities	14,02	Required
6	Dr. Yoo Taek Lee	Executive Dean of International Schools (Solbridge, Endicott & JCFS) and Dean of Endicott College & JWKIM College of Future Studies. For more than 20 years he worked at Boston University as an Operation Professor (recognized as the best professor by student surveys). Author of a few books in Operation Management. Helped in building operation Management in multinational corporations such as GAP, Samsung and others.	Supply Chain Basic	9,2	Required
7	Dr. Yoo Taek Lee	Executive Dean of International Schools (Solbridge, Endicott & JCFS) and Dean of Endicott College & JWKIM College of Future Studies. For more than 20 years he worked at Boston University as an Operation Professor (recognized as the best professor by student surveys). Author of a few books in Operation Management. Helped in building operation	Matching Supply and Demand	11,2	Required

		Management in multinational corporations such as GAP, Samsung and others.			
8	Dr. Yoo Taek Lee	Executive Dean of International Schools (Solbridge, Endicott & JCFS) and Dean of Endicott College & JWKIM College of Future Studies. For more than 20 years he worked at Boston University as an Operation Professor (recognized as the best professor by student surveys). Author of a few books in Operation Management. Helped in building operation Management in multinational corporations such as GAP, Samsung and others.	Inventory Management	17,16	Required
9	Dr. Yoo Taek Lee	Executive Dean of International Schools (Solbridge, Endicott & JCFS) and Dean of Endicott College & JWKIM College of Future Studies. For more than 20 years he worked at Boston University as an Operation Professor (recognized as the best professor by student surveys). Author of a few books in Operation Management. Helped in building operation Management in multinational corporations such as GAP, Samsung and others.	Various Types of Supply	20,23	Required
10	Dr. Yoo Taek Lee	Executive Dean of International Schools (Solbridge, Endicott & JCFS) and Dean of Endicott College & JWKIM College of Future Studies. For more than 20 years he worked at Boston University as an Operation Professor (recognized as the best professor by student surveys). Author of a few books in Operation Management. Helped in building operation Management in multinational corporations such as GAP, Samsung and others.	Special Session 1: Socially Responsible Supply Chains	9,44	Required
11	Dr. Yoo Taek Lee	Executive Dean of International Schools (Solbridge, Endicott & JCFS) and Dean of Endicott College & JWKIM College of Future Studies. For more than 20 years he worked at Boston University as an Operation Professor (recognized as the best professor by student surveys). Author of a few books in Operation Management. Helped in building operation Management in multinational corporations such as GAP, Samsung and others.	Special Session 2: Quality Introduction	19,02	Required
12	Dr. Yoo Taek Lee	Executive Dean of International Schools (Solbridge, Endicott & JCFS) and Dean of Endicott College & JWKIM College of Future Studies. For more than 20 years he worked at Boston University as an Operation Professor (recognized as the best professor by student surveys). Author of a few books in Operation Management. Helped in building operation Management in multinational corporations such as GAP, Samsung and others.	Special Session 3: Cost of Quality	19,48	Required
13	Joleen Liang	Co-Founder & International CEO, Squirrel Ai Learning, a Chinese Unicorn, with over 60,000+ full-time primary and secondary schools. Squirrel Ai Learning's smart education initiatives earned the 'AI Education Innovation Award' from UNESCO in 2020 and recognition in the inaugural edition of TIME Magazine's World's Top EdTech Companies 2024. China's Power Business Women by Forbes in 2025. Joleen also holds position as an Executive director of the AI and Robotics Education Committee of the China Education Development Strategy Society	Revolutionizing personalized learning through AI	76,17	(required /optional depends on chosen 4 industries)
14	Dr. Eva Coates	Founding Partner of Clinical AI Partners Oxford. Consultant Paediatric Surgeon with 20 years of National Healthcare Service in the UK and one of the UK's leading voices on Clinical AI education. She works with NHS Trusts and global health partners to bridge the AI fluency gap - helping clinicians work safely and effectively alongside AI tools. Her platform, Lyra, spans 11 product lines from hackathons and micro-learning to workforce audit tools and regulatory training.	AI in Healthcare Transformation	78,41	(required /optional depends on chosen 4 industries)

15	Pradeep Menon	Serial entrepreneur in Cybersecurity and AI. Co-Founder and CEO at InceptAI. Chief AI and Information Security Officer (CAISO) at Paramount Computer Systems. Member oa AI for developing countries. Director at Cybersecurity-NxxT. Adjunct Professor of Practice at School of Artificial Intelligence of Amrita Vishwa Vidyapeetham	How AI transforming Cybersecurity in Telecom	80,47	(required /optional depends on chosen 4 industries)
16	Gerald Perez	CEO of Interactive Brokers, widely recognized as one of the most advanced and respected brokerage firms in the global market, with a market capitalization of around 100 billion dollars	How AI transforms Financial Services	81,15	(required /optional depends on chosen 4 industries)
17	Kevin Dowling	Managing Director of the Robotics Factory and Hardware at Innovation Works - Pittsburgh's flagship startup program, part of a $63M U.S. Economic Development Administration grant to grow the robotics and AI startup ecosystem across the region that is widely recognized as the "Robotics Capital of the World," with 100+ robotics companies clustered there. Kevin leads the Factory's three interconnected programs - Create, Accelerate, and Scale - which work specifically with sectors including construction, aerospace, mining, and defense to identify industry problems and turn them into venture-backed robotics startups. Also a founding member and Board Chair of the Pittsburgh Robotics Network, the regional industry body coordinating talent, capital, and policy for the sector.Holds a PhD and spent 17 years as Scientist and Senior Engineer at Carnegie Mellon University's Robotics Institute, one of the most influential robotics research centers in the world. Former VP of Color Kinetics (acquired by Philips), where his work on intelligent LED lighting systems generated one of the most valuable patent portfolios in the industry.	Construction Market Map and Robotics Case Study	35,35	(required /optional depends on chosen 4 industries)
18	Sanzhar Rakhmetzhanov	Co-Founder & Chief Engineering Intelligence Officer at Armeta, Alchemist graduate, a winner at the Energy Venture Day&Pitch Competition by S&P 500	How AI transforms legal documents into Profitable Data in Industrial Construction	81,1	(required /optional depends on chosen 4 industries)
19	Joe Ward	Account Director at Snowflake. One of the world's leading data and AI platform companies, which helps organizations bring together their data, systems, and teams so they can use data and AI in a much more connected and effective way. Snowflake's market capitalization is about $57.49 billion, Warren Baffet invested around Snowflake's IPO in 2020 and Reuters described Snowflake's debut as the largest U.S. listing of that year.	How Snowflake turns data into AI Strategy	87,55	(required /optional depends on chosen 4 industries)
20	Chi Chiu	Founder of CosmicBrain AI. (Building smarter robots for the future with AI), Ex serial Intrapreneur & Technical Leader with Global Operations Expertise at Google, Inc. with over 25 years of experience building successful ventures worldwide and leading multiple Google products from incubation to scaling, working directly with Sergey Brin and Larry Page	Robots in Mining	71,19	(required /optional depends on chosen 4 industries)
21	Mahi de Silva	Co-Founder and Chief Strategy Officer at Higgsfield. Ex executive Vice President, Consumer Mobile Opera Software ASA. Ex Chief Executive Officer Triller. Board Member and Advisor in several startups. Silicon Valley veteran with deep experience building, scaling, and commercializing technology companies.	Higgsfield AI Democratizes Content Creators for Social Media Marketers	27,14	(required /optional depends on chosen 4 industries)
22	James D. Boyd	Chief Information Officer and Vice President, Digital Technology & Solutions at Valeura Energy (one of largest energy company with HQ in Canada) in the Asia-Pacific region. Principal Consultant, Global CIO Advisor in Digital Transformation, AI & Cyber Infrastructure.	AI in Energy Sector	58,58	(required /optional depends on chosen 4 industries)

23	Luca Albertalli	Senior Director of Product Management at Guild, an unicorn, has been named on Fortunes ChangetheWorld list. Co-Founded the Theneeds, Inc. Acquired by shopkick, Inc.	AI is reshaping talent management	84,03	(required /optional depends on chosen 4 industries)
24	Irakli Chikava	Co-founder and CEO of Telagri - an agri-fintech platform helping banks make smarter agricultural lending decisions through real-time farm performance and risk insights. Based in Tbilisi, Georgia, operating across Eastern Europe and Central Asia Startup 15+ years in agriculture and agribusiness consulting, with involvement in $40M+ of investment projects across emerging markets. World Cup Finalist, Speaker, Agritech Professional. Co-organizer of Future Agro Challenge Georgia. Co-Founder and Commercial Director at Agro Solutions Group	De-risking Agricultural Lending in Emerging Markets	71,04	(required /optional depends on chosen 4 industries)
Subtotal Module 2.2 Industry Knowledge (minutes)				**989,36**	
Module 2.3 Trends and Case Studies in Local and International Markets					
1	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	AI & Industry: Unlocking Global Opportunities from Central Asia	23,38	Required
2	Nick Kelley	Partner at Alpha Intelligence Capital. Ex-Director Data Science & A.I. Innovation for the Novartis Digital Office. A.I. Strategy and Deep Tech Advisor at Blue Zebra. Collaborating with MIT and Microsoft Research	Healthcare & Life Scince Sector overview and AI outlook	75,18	Required
3	Aset Abdualiev	Founder & CEO at Silkroad Innovation Hub. Founder of Stanford Central Asia Hub Startup Mentor at Alchemist Accelerator Founding Partner at BuildTech VC Board Trustee at American University of Central Asia (AUCA)	Central Asia's Startups are Gaining Access to Silicon Valley	42,47	Optional
4	Magzhan Madiyev	Founding CEO of Astana Hub. Co-Founder of Naimi.kz. Member of the Board of Directors, Independent Director at Astana Innovations, Republican Center for Space Communication, Member of the Board of Directors at Grand Mobile, Axellero.io	From Local to Global: International Markets with Astana Hub	33,36	Optional
5	Alex Aguilar	Associate and Team Lead for Mining at Mind the Bridge a global open innovation platform headquartered in Silicon Valley that connects Fortune 500 corporates with startups. Leads the mining innovation vertical, helping companies in mining, defense, shipbuilding, and energy identify and integrate emerging technologies. Over 8 years of experience in business development and innovation across LATAM, Asia, and Europe. Previously led B2B business development at Huawei in Latin America and held senior roles at Linio, Sodimac, and Grupo Pacific Traders. At Mind the Bridge has supported Fortune 500 energy and cleantech corporates - including Enel, ENI, and Terna - with startup scouting, M&A sourcing, benchmarking, and market landscaping. Holds an executive education at MIT.	Minerals and Mining Market Map	69,24	Required
6	Michael Mo	Founder and CEO at GSV Holdings. Co-founder of the ASU GSV Summit. The world's largest event at the intersection of education, edtech, workforce skills, and venture capital. Forbes calls it the "Davos of Education," and The New York Times describes it as the "must-attend event for education technology." Founded in 2010 in partnership with Arizona State University and GSV Ventures (Global Silicon Valley). Founder of GSV	Education Market Map	44,13	Required

		University and GSV MBA. Author of books: “Finding the Next Starbucks”, “The Global Silicon Valley Handbook,” “The Mission Corporation...Contemporary Capitalism”			
7	Guilianna Crivello	Founder of Stealth Startup, a network for entrepreneurs building in stealth and a venture partner at OFF PISTE, a venture studio operating at the intersection of thematic investing, control/ownership models, and vertical AI implementation across private equity and startups. Previously Founding General Partner and Head of Ventures at Draper Startup House Ventures (part of the Draper Venture Network, - launched the firm's first global micro-fund in 2021 together with Vikram Bharati, investing in early-stage founders across 60+ countries and 30+ industries, with strong exposure to fintech deal flow in emerging markets across Southeast Asia, India, Latin America, and Eastern Europe. Made 200+ investor introductions to global founders and led the firm's transition from a hospitality-first startup platform into a true cross-border venture fund during the pandemic	Fintech DFAI Market Map	66,04	Optional
8	Tahira Dosani	General Partner and Co-Founder of ResilienceVC. Adjunct Professor at The Johns Hopkins University - Paul H. Nitze School of Advanced International Studies (SAIS). Adjunct Professor at Georgetown University	Fintech Market Map	49,23	Required
9	Axel Badalian	Principal at Alpha Intelligence Capital (AIC) - a $500M global venture capital fund dedicated exclusively to deep-tech AI and machine-learning companies, investing from Seed to Series B with a globally distributed team of deep-tech entrepreneurs, scientists, and investors. Based in Paris, Axel leads sourcing and investment execution across the AI/ML enabling stack - from compute, hardware, and infrastructure to foundation models and verticalized applications. He is also listed as an AI Expert at OECD.AI, contributing to policy discussions on AI computing capacity, infrastructure scale-up, and its environmental impact.	Computing Infrastructure with VC	78,42	Required
10	Arun Vijay	Investment Analyst at Alpha Intelligence Capital (AIC) - a $500M global venture capital fund dedicated exclusively to deep-tech AI and machine-learning companies, investing from Seed to Series B out of offices in San Francisco and Singapore. At AIC, Arun supports sourcing and due diligence across the AI/ML enabling stack, with direct exposure to the cybersecurity portfolio - a core thesis area for the fund, which includes investments in ZeroEyes (AI-powered gun detection for public and private spaces), X-37 (AI-driven cyber defense), and infrastructure-security plays like Exostellar (secure AI workload orchestration across heterogeneous hardware - acquired by Qualcomm). These investments sit at the center of the defining cybersecurity questions of the AI era: how to defend AI systems themselves, how to use AI to detect and respond to threats at machine speed, and how to secure the compute and data infrastructure on which modern enterprises now depend..In parallel, he is building OpenMentor.AI, a community-driven mentorship platform for students.	Cybersecurity Market Map	56,14	Optional
11	David Forsberg	Managing Partner and Founder of Ascent Energy Ventures - a Denver-based specialist venture capital firm focused on the digital transformation of the energy industry. Ascent invests in startups building the software, automation, AI, and data infrastructure layer of the global energy stack - from modernizing legacy SCADA systems in oil & gas, to regulatory intelligence tooling, grid-planning software, and digital platforms across power, utilities, and hydrocarbons. Ascent's thesis is explicitly pragmatic: rather than picking a side between hydrocarbons and decarbonization, the firm backs the founders building the automation and digital tools that make all forms of	Digital and Energy Sector	63,42	required

		energy production more efficient, transparent, and measurable. Also serves as a strategic advisor to energy founders and private investors			
12	Charlie Kleinsmith	Principal at SineWave Ventures - a $160M early-stage venture capital firm with offices in New York and San Francisco, dedicated to backing technologies that bridge commercial enterprise and public-sector adoption. Charlie leads investments at the intersection of data, AI, cybersecurity, networking, and compute infrastructure - precisely the stack underpinning modern operational automation. Publishes thought leadership on data provenance, zero-trust security architectures, and data control frameworks, which are central to deploying automation inside regulated enterprises and government agencies. SineWave's partnership bench itself - former leaders from NIST (cybersecurity program lead), CIA, NSA, Department of Commerce, GSA, and DoD.	Operational Automation Market Map	57,28	Optional
13	Meïssa Gueye	Head of Venture Capital Africa at IFC (International Finance Corporation) - a member of the World Bank Group and the largest global development institution focused exclusively on the private sector in emerging markets, with a record $71.7 billion committed in fiscal year 2025. Meïssa leads IFC's equity and fund investments into African tech and growth-stage companies, deploying capital directly and via VC funds and accelerators across Africa. IFC is the largest emerging-market fintech investor globally, with an early-stage fintech portfolio exceeding $600M and over $3B committed across 175+ tech companies and 120+ VC funds. Based in Dakar, Meïssa sits at the center of the African venture ecosystem - frequently cited as one of the investors every African founder should know.	Entertainment and Media Market Map	62,49	Optional
14	Kieran Gartlan	Managing Partner at the Yield Lab LATAM, one of Latin America's leading AgriFood and Climate Tech venture capital firms, with offices in Argentina, Brazil, and Chile, and part of The Yield Lab global network with regional funds across North America, Europe, Latin America, and Asia Pacific. Kieran leads a portfolio spanning the full farm-to-fork spectrum - with access to 1,100+ startups across Latin America. Board Member at Produzindo Certo, BemAgro S.A, Grão Direto, SEEDZ. Winner of Jesse H Neal Award 2010 for best recurring Webcast covering South American Grains Market. Winner of National Excellence Award 2007 from the NAAJ (North American Agricultural Journalists Association) for analysis of Brazil’s grain sector. Author of "Cereal Killers – Grain Markets 101” commissioned by U.K publisher Harriman House	Food and Agriculture Market Map	55,58	Required
15	Kevin Dowling	Managing Director of the Robotics Factory and Hardware at Innovation Works - Pittsburgh's flagship startup program, part of a $63M U.S. Economic Development Administration grant to grow the robotics and AI startup ecosystem across the region that is widely recognized as the "Robotics Capital of the World," with 100+ robotics companies clustered there. Kevin leads the Factory's three interconnected programs - Create, Accelerate, and Scale - which work specifically with sectors including construction, aerospace, mining, and defense to identify industry problems and turn them into venture-backed robotics startups. Also a founding member and Board Chair of the Pittsburgh Robotics Network, the regional industry body coordinating talent, capital, and policy for the sector.Holds a PhD and spent 17 years as Scientist and Senior Engineer at Carnegie Mellon University's Robotics Institute, one of the most influential robotics research centers in the world. Former VP of Color Kinetics (acquired by Philips), where his work on intelligent	Robotics Market Map	65,15	Required

		LED lighting systems generated one of the most valuable patent portfolios in the industry.			
16	Askar Aituov	Founder of Devs.kz. Ex-Google Community Manager. Senior Lecturer and a PhD Candidate in Information Systems at Kazakh-British Technical University. Managed the Silkway Accelerator (Google for Startups). Ex Developer and Startup relations for Google developers ecosystem (via EPAM)	Construction Market Map and Robotics Case Study	34,37	Optional
Subtotal Module 2.3 Trends and Case Studies in Local and International Markets (minutes)				**875,88**	
Module 2.4 Hard Entrepreneurial Skills					
1	Bola Adegbulu	AI Consultant at Infinite Ascent Ventures.Ex-Principal in AI Fund and acceleration program, led by Prof. Andrew Ng's. Board Member at Africa Deep Tech Community. Member of PEF Community. Europe 30 under 30: Manufacturing & Industry.	Transforming AI Prototypes	16,12	Required
2	Bola Adegbulu	AI Consultant at Infinite Ascent Ventures.Ex-Principal in AI Fund and acceleration program, led by Prof. Andrew Ng's. Board Member at Africa Deep Tech Community. Member of PEF Community. Europe 30 under 30: Manufacturing & Industry.	The Filter (part 1)	17,13	Required
3	Bola Adegbulu	AI Consultant at Infinite Ascent Ventures.Ex-Principal in AI Fund and acceleration program, led by Prof. Andrew Ng's. Board Member at Africa Deep Tech Community. Member of PEF Community. Europe 30 under 30: Manufacturing & Industry.	The Filter (part 2)	16,37	Required
4	Bola Adegbulu	AI Consultant at Infinite Ascent Ventures.Ex-Principal in AI Fund and acceleration program, led by Prof. Andrew Ng's. Board Member at Africa Deep Tech Community. Member of PEF Community. Europe 30 under 30: Manufacturing & Industry.	Sourcing Ideas	12,33	Required
5	Bola Adegbulu	AI Consultant at Infinite Ascent Ventures.Ex-Principal in AI Fund and acceleration program, led by Prof. Andrew Ng's. Board Member at Africa Deep Tech Community. Member of PEF Community. Europe 30 under 30: Manufacturing & Industry.	Technical Evaluation	18,48	Required
6	Bola Adegbulu	AI Consultant at Infinite Ascent Ventures.Ex-Principal in AI Fund and acceleration program, led by Prof. Andrew Ng's. Board Member at Africa Deep Tech Community. Member of PEF Community. Europe 30 under 30: Manufacturing & Industry.	Step one: Building the prototype	19,02	Required
7	Bola Adegbulu	AI Consultant at Infinite Ascent Ventures.Ex-Principal in AI Fund and acceleration program, led by Prof. Andrew Ng's. Board Member at Africa Deep Tech Community. Member of PEF Community. Europe 30 under 30: Manufacturing & Industry.	Step Two: Ready to Build	14,13	Required
8	Bola Adegbulu	AI Consultant at Infinite Ascent Ventures.Ex-Principal in AI Fund and acceleration program, led by Prof. Andrew Ng's. Board Member at Africa Deep Tech Community. Member of PEF Community. Europe 30 under 30: Manufacturing & Industry.	Design Partner	34,49	Required
9	Bola Adegbulu	AI Consultant at Infinite Ascent Ventures.Ex-Principal in AI Fund and acceleration program, led by Prof. Andrew Ng's. Board Member at Africa Deep Tech Community. Member of PEF Community. Europe 30 under 30: Manufacturing & Industry.	MVP to Production Solution	27,49	Required
10	Bola Adegbulu	AI Consultant at Infinite Ascent Ventures.Ex-Principal in AI Fund and acceleration program, led by Prof. Andrew Ng's. Board Member at Africa Deep Tech Community. Member of PEF Community. Europe 30 under 30: Manufacturing & Industry.	Market Sizing (TAM, SAM, SOM)	18,43	Required
11	Bola Adegbulu	AI Consultant at Infinite Ascent Ventures.Ex-Principal in AI Fund and acceleration program, led by Prof. Andrew Ng's. Board Member at Africa Deep Tech Community. Member of PEF Community. Europe 30 under 30: Manufacturing & Industry.	Competitive Landscape	24,01	Required
12	Bola Adegbulu	AI Consultant at Infinite Ascent Ventures.Ex-Principal in AI Fund and acceleration program, led by Prof. Andrew Ng's. Board	Customer Segmentation	26,59	Required

		Member at Africa Deep Tech Community. Member of PEF Community. Europe 30 under 30: Manufacturing & Industry.			
13	Bola Adegbulu	AI Consultant at Infinite Ascent Ventures.Ex-Principal in AI Fund and acceleration program, led by Prof. Andrew Ng's. Board Member at Africa Deep Tech Community. Member of PEF Community. Europe 30 under 30: Manufacturing & Industry.	Lean Startup Metodology	12,55	Required
14	Bola Adegbulu	AI Consultant at Infinite Ascent Ventures.Ex-Principal in AI Fund and acceleration program, led by Prof. Andrew Ng's. Board Member at Africa Deep Tech Community. Member of PEF Community. Europe 30 under 30: Manufacturing & Industry.	Matching your Launch to Aproach Costumer	13,25	Required
15	Bola Adegbulu	AI Consultant at Infinite Ascent Ventures.Ex-Principal in AI Fund and acceleration program, led by Prof. Andrew Ng's. Board Member at Africa Deep Tech Community. Member of PEF Community. Europe 30 under 30: Manufacturing & Industry.	The AI Cost Sctucture	19,21	Required
16	Bola Adegbulu	AI Consultant at Infinite Ascent Ventures.Ex-Principal in AI Fund and acceleration program, led by Prof. Andrew Ng's. Board Member at Africa Deep Tech Community. Member of PEF Community. Europe 30 under 30: Manufacturing & Industry.	Unit economics and margin	12,51	Required
17	Bola Adegbulu	AI Consultant at Infinite Ascent Ventures.Ex-Principal in AI Fund and acceleration program, led by Prof. Andrew Ng's. Board Member at Africa Deep Tech Community. Member of PEF Community. Europe 30 under 30: Manufacturing & Industry.	AI Pricing Models	13,05	Required
18	Bola Adegbulu	AI Consultant at Infinite Ascent Ventures.Ex-Principal in AI Fund and acceleration program, led by Prof. Andrew Ng's. Board Member at Africa Deep Tech Community. Member of PEF Community. Europe 30 under 30: Manufacturing & Industry.	Building a Financial Model that works	18,01	Required
19	Bola Adegbulu	AI Consultant at Infinite Ascent Ventures.Ex-Principal in AI Fund and acceleration program, led by Prof. Andrew Ng's. Board Member at Africa Deep Tech Community. Member of PEF Community. Europe 30 under 30: Manufacturing & Industry.	Selling what you've build	15,08	Required
20	Bola Adegbulu	AI Consultant at Infinite Ascent Ventures.Ex-Principal in AI Fund and acceleration program, led by Prof. Andrew Ng's. Board Member at Africa Deep Tech Community. Member of PEF Community. Europe 30 under 30: Manufacturing & Industry.	Finding your first costumer	17,41	Required
21	Bola Adegbulu	AI Consultant at Infinite Ascent Ventures.Ex-Principal in AI Fund and acceleration program, led by Prof. Andrew Ng's. Board Member at Africa Deep Tech Community. Member of PEF Community. Europe 30 under 30: Manufacturing & Industry.	Building your simple revenue engine	18,57	Required
22	Bola Adegbulu	AI Consultant at Infinite Ascent Ventures.Ex-Principal in AI Fund and acceleration program, led by Prof. Andrew Ng's. Board Member at Africa Deep Tech Community. Member of PEF Community. Europe 30 under 30: Manufacturing & Industry.	Scaling the go to market motion	18,48	Required
23	Peter Turner	Partner at Breakthrough Energy Ventures. BEV invests in companies that leverage innovative technologies to help address climate change. Backed by many of the world's top business leaders, BEV has more than $2 billion in committed capital to support bold entrepreneurs building companies that can significantly reduce emissions from agriculture, buildings, electricity, manufacturing, and transportation. EVP Selection Committee in Stanford Woods Institute for the Environment	Food & Agriculture: Analyzing the Industry, Building Go To Market, and Winning Over VCs	55,43	Optional
24	Bolat Sultankulov	Founder, CEO at ARLAN BIOTECH. StartX S24 cohort Co-Founder, CEO at Astana Biomed Group. Founder, CEO at Kletochnaya terapiya. Doctor of Philosophy - PhD, Bioengineering and Biomedical Engineering at Nazarbayev University.	Biotech & Life Sciences: How to Enter Deep Tech, Build a Small AI-Leveraged Team, and Win Over Top Accelerators	52,33	Optional

25	Seamus McAteer	Chief Executive Officer at Speechlab. Non Executive Director at On Device Research. Founder in residence in AI fund (Managed by Andrew Ng). Investor & Advisor. 3 exits.	AI Applications: How to Define Your Ideal Customer Profile, Build Moats Against Foundational Players, and Raise Capital Judiciously	42,11	Optional
26	Alberto Pellicione	Partner at Alpha Intelligence Capital. AI Investor. Ex Global XDR Strategy Leader at IBM, participated in developing QRadar+ XDR strategy on a global scale. Serial Entrepreneur in Cybersecurity field.	AI Products in Cybersecurity: How to Start from the Problem, Build Iteratively, and Defend Against Foundation Models	86,52	Optional
27	Hyu Dong	NUS Senior Sales Specialist at Nutanix, a global leader in cloud software, offering organizations a single platform for running apps and data across clouds, with capitalisation more than 10 bln dollars. 3 IPO/Acquisitions	Sales in B2B Enterprise Tech: How to Sell Before You Have a Product, Run the Full Sales Funnel, and Hire Your First Sales Team	44,65	Optional
28	Talgat Kantayev	Co-founder of InvestBanq (Singapore). , Advisor at Bueno.money With more than 8 years in VC (Paladigm Capital,Eurasia RED)	Fintech & Wealth Management: How to Build AI Products in a Trust-Based Industry, Segment High-Net-Wealth Clients, and Scale Cross-Cultural Teams	52,19	Optional
29	Tom Pereira	Chief Financial Officer at at Advanced Navigation, an Australian unicorn, a global leader in navigation and autonomous systems which serves over 2800 clients across 70+ countries (among clients Boeing, Airbus, NASA and others)	Startup Finance in Tech: How to Master Financial Statements, Understand CAC-to-LTV, and Balance Growth vs. Profitability	63,43	Optional
30	Scott Nuzum, Scott Felder	Scott Nuzum - Special Counsel at Wiley Rein Attorney at Wiley Rein LLP (Washington, D.C.) - advising dual-use and deep tech companies on corporate law, national security, and emerging technology regulation: M&A, cybersecurity compliance, and technology licensing. Works with founders and venture capital funds across the defense, energy, space, and AI sectors on cross-border deals. Over 15 years of experience at the intersection of law, business, and public policy. Previously held legal and policy positions in the US federal government, served as General Manager at the venture studio Innovyz USA, co-founded a data marketplace for agtech applications. Scott Felder - Partner at Wiley Rein LLP (Washington, D.C.) and head of the firm's Patent Prosecution practice. Focuses on complex IP issues faced by government contractors, including patent rights, rights in technical data, and computer software, with emphasis on the aerospace, defense, and intelligence sectors. Represents clients in IP disputes and negotiations with the US government. Named one of DC's "Super Lawyers" for Intellectual Property multiple times (2013, 2015–2020, 2023). Frequent speaker and author on IP, and co-instructor of the Federal Publications Seminars Government Contract Intellectual Property Workshop and the Public Contracting Institute's Demystifying Intellectual Property in Government Contracting series.	Startup Legal for AI Companies: How to Architect Your Company, Structure for Global VC Investment, and Protect IP in the AI Era	66,53	Optional

31	Brady Forest	Founder of Ignite Talks, a global talk series. oard Member, x-Head of Business Development at Sentz Global.Board Member at Ideas Beyond Bordersand Community Arts Stabilization Trust Guest Host and Curator at TED Conferences. Ex - Venture Partner, GenLab Venture Studio	Communication for Founders in the AI Era: How to Know Your Audience, Tell Stories That Land, and Use AI Without Losing Your Voice	48,09	Optional
Subtotal Module 2.4 Hard Entrepreneurial Skills (minutes)				**913,99**	
Module 2.5 Soft Skills for Leadership and Teamwork					
1	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Effective communication fundamentals	19,19	Required
2	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Cross-cultural communication	24,31	Required
3	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Persuasion and influence	23,38	Required
4	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Storytelling for business	16,25	Required
5	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Presenting with Confidence	24,2	Required
6	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have	Negotiation Skills for Entrepreneurs	16,17	Required

		been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank			
7	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Internal team communication	16,16	Required
8	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Facilitation mastery	18,14	Required
9	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Building high-performing teams	16,16	Required
10	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Team dynamics and roles	20,48	Required
11	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Giving and receiving feedback	19,23	Required
12	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Collaborative problem-solving	15,46	Required
13	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile	Conflict resolution in teams	18,09	Required

		Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank			
14	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Building team culture	19,24	Required
15	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Effective delegation	20,03	Required
16	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Cross-functional collaboration	21,16	Required
17	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Remote and Hybrid Team Leadership	12,5	Required
18	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Intergenerational Leadership	13,33	Required
19	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Team motivation and morale	19,12	Required

20	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	The human side of hiring	17,01	Required
21	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Leadership vs. management	14,41	Optional
22	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Decision-making under uncertainty	21,42	Optional
23	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Emotional intelligence for leaders	21,4	Optional
24	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Servant leadership	20,54	Optional
25	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Ethical leadership and values-driven decision making	12,42	Optional
26	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean	Leading with data	17,51	Optional

		and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank			
27	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Leading in the age of AI	18,5	Optional
28	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Crisis leadership	18,35	Optional
29	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Managing Up and Across	12,11	Optional
30	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Building strategic partnerships and alliances	18,59	Optional
31	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Sales psychology for founders	11,32	Optional
32	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Adaptability and pivoting	13,11	Optional
33	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco	Critical thinking skills	22,46	Optional

		described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank			
34	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Creative problem-solving	22,06	Optional
35	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Time management and productivity	22	Optional
36	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Managing your energy	14,39	Optional
37	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Stress management and resilience	24,3	Optional
38	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Growth mindset cultivation	21,52	Optional
39	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Self-awareness and reflection	19,48	Optional

40	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Goal setting and achievement	23,31	Optional
41	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Learning how to learn	20,11	Optional
42	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Building professional network	18,56	Optional
43	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Personal branding	18,38	Optional
44	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Financial literacy for founders	14,17	Optional
45	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Health and wellness for entrepreneurs	17,38	Optional
46	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean	Continuous learning plan	18,41	Optional

		and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank			
Subtotal Module 2.5 Soft Skills for Leadership and Teamwork (minutes)				**845,82**	
Module 2.6 The final stage					
1	Paul Kim	PhD - Advisor in Lumos Capital Group. Member of National Artificial Intelligence Council President of the Republic of Kazakhstan. He is the creator of SMILE (Stanford Mobile Inquiry Learning Environment), a platform that Unesco described as a paradigm shift in education. Presidents of Seeds of Empowerment, an international initiative whose projects have been implemented in more than 22 countries. Ex Associate Dean and CTO at Stanford University Graduate School of Education. Ex-Chair of International Expert Committee of World Bank	Your capstone project pitch and presentation overview	33,08	Required
Subtotal Module 2.6 The final stage (minutes)				33,08	
TOTAL time (minutes)				**4496,46**	